



NORTH SYDNEY GIRLS HIGH SCHOOL
HSC Extension 1 Mathematics Assessment Task 1
Term 4, 2013

Name: _____ **Mathematics Class: 11Mx** _____

Student Number: _____

Time Allowed: 55 minutes + 2 minutes reading time

Total Marks: 42

Instructions:

- Attempt all questions.
- Start each question in a new booklet. Put your name on every booklet.
- Show all necessary working.
- Marks may be deducted for incomplete or poorly arranged work.
- Work down the page.
- Do not work in columns.
- Each question will be collected separately. Submit a blank booklet if you do not attempt a question.

Question	1	2	3-5	6	7	8 a	8 b,c	Total
PE3	/1			/6	/3		/7	/17
PE4		/1		/7				/8
HE7			/3		/9	/5		/17
	/1	/1	/3	/13	/12	/5	/7	/42

Section I

5 marks

Attempt Questions 1 - 5

Use the multiple choice answer sheet for Questions 1–5

- 1 One of the factors of $P(x) = ax^3 - 7x^2 + kx + 4$ is $(x - 4)$ and the remainder when $P(x)$ is divided by $(x - 1)$ is -6 .
- Which of the following is correct?
- (A) $16a + k = -27$ and $a + k = -3$
- (B) $16a + k = 27$ and $a + k = 3$
- (C) $16a + k = -27$ and $a + k = 3$
- (D) $16a + k = 27$ and $a + k = -3$
- 2 Two points $P(2t, t^2)$ and $Q(2s, s^2)$ lie on the parabola $x^2 = 4y$. It is known that $ts = -4$. What are the coordinates of the midpoint of PQ ?
- A) $\left(\frac{-8}{t}, \frac{16}{t^2}\right)$
- B) $\left(\frac{t^2 - 4}{t}, \frac{t^4 + 16}{2t^2}\right)$
- C) $\left(-t, \frac{17t^2}{2}\right)$
- D) $\left(\frac{2t^2 - 4}{t}, \frac{t^4 + 16}{t^2}\right)$
- 3 The derivative of a function is given as $f'(x) = \frac{x^2}{x - 2}$. Which of the following could be true of the original function $y = f(x)$ at the point where $x = 0$?
- A) There is a horizontal point of inflection
- B) There is a minimum turning point
- C) There is a maximum turning point
- D) The gradient is not defined

- 4 The chord of contact from an external point $A(x_0, y_0)$ to the general parabola $x^2 = 4ay$ has equation $xx_0 = 2a(y + y_0)$. From what external point are the tangents to the parabola $x^2 = 6y$ to be drawn so that $2x - 3y - 3 = 0$ is the chord of contact?

- A) $(1, 1)$
- B) $(2, 1)$
- C) $(1, -3)$
- D) $(2, -3)$

- 5 What is the derivative of $\frac{x}{\sqrt{1-2x}}$?

- A) $\frac{1-x}{(1-2x)\sqrt{1-2x}}$
- B) $\frac{1+x}{\sqrt{1-2x}}$
- C) $\frac{1-3x}{(1-2x)\sqrt{1-2x}}$
- D) $\frac{1+x-2x^2}{\sqrt{1-2x}}$

Section II

38 marks

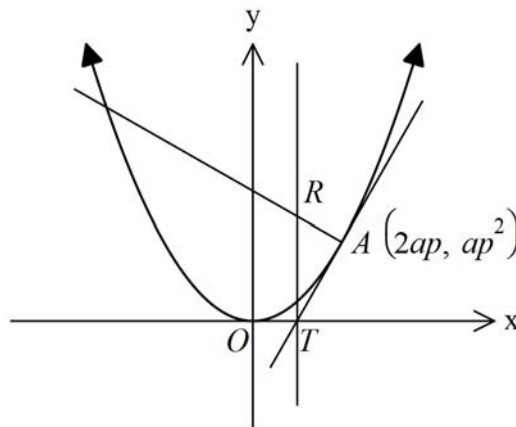
Attempt Questions 6–8

Answer each question in a SEPARATE writing booklet. Extra writing paper is available.

In Questions 6–8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (13 marks) Use a SEPARATE writing booklet.

- a) By writing down factors, or otherwise, construct a monic polynomial of the form $x^3 + bx^2 + cx + d$ which has zeros $1, 2 + \sqrt{2}, 2 - \sqrt{2}$. 3
- b) Write $x^3 - x^2 - 4x$ in the form $A(x-1)^3 + B(x-1)^2 + C$ 3
- c) The parabola has parametric equations $x = 2at$ and $y = at^2$. A tangent and normal have been drawn at a variable point $A(2ap, ap^2)$ which lies on the parabola.



- (i) Show that the equation of the tangent at the point A is $y = px - ap^2$ 2
- (ii) The tangent cuts the x axis at T . Find the coordinates of T . 1
- (iii) The equation of the normal at A is $x + py = 2ap + ap^3$. Do NOT show this.
A line through T parallel to the axis of the parabola cuts the normal at R . Show that the coordinates of R are $(ap, ap^2 + a)$. 1
- (iv) Show that the locus of R is a parabola and state the equation of its directrix. 3

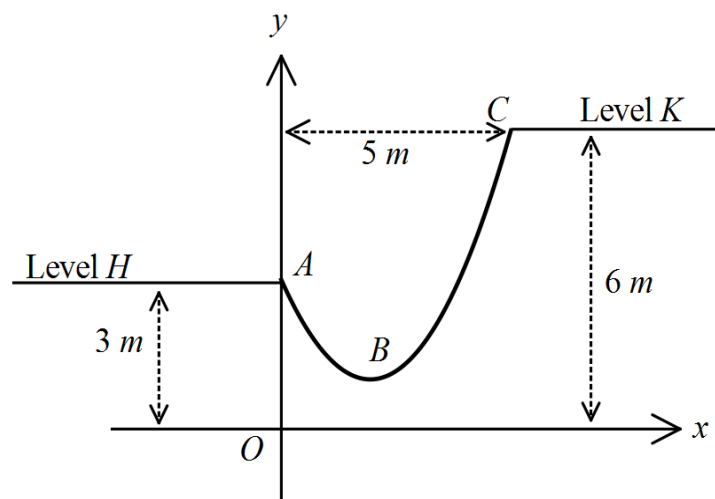
Question 7 (12 marks) Use a SEPARATE booklet.

- a) Solve $x^3 - 2x^2 - 7x + 2 = 0$. **3**
- b) Let $P(x) = -2x^3 + kx^2 - mx + 5$. Show that if $P(x)$ is to have any stationary points, then $k^2 - 6m \geq 0$. **2**
- c) Consider the curve given by the equation $y = \frac{(x-1)^2}{x+1}$.
- (i) Show that there are stationary points at $x = -3$ and $x = 1$, and determine their nature. **3**
- (ii) Show that $\frac{(x-1)^2}{x+1} = x - 3 + \frac{4}{x+1}$. **1**
- (iii) Sketch the curve showing all important features. **3**

Question 8 (12 marks) Use a SEPARATE booklet

- a) The city council has decided to build a skateboard ramp for its teenagers. The structure will consist of two levels, H and K and a ramp, as shown in the diagram. The engineers believe that if the ramp has a gradient of greater than 3 at any point, it will be too dangerous to use.

A cross-section of the proposed ramp is shown below.



The ramp ABC is given by the equation $y = \frac{8x^2}{15} - \frac{31x}{15} + 3$ for $0 \leq x \leq 5$.

- | | | |
|-------|---|---|
| (i) | Use the information given in the cross-section to write down the coordinates of the point C . The point A is $(0, 3)$. | 1 |
| (ii) | Determine the maximum gradient of the ramp over this domain. | 2 |
| (iii) | What is the greatest height that level K may be constructed so that the ramp is deemed safe? Give your answer correct to 1 decimal place. | 2 |
- b) The parametric equations of a curve are $x = t^2 + 1$ and $y = \frac{1}{t}$.
- Without eliminating t ,
- | | | |
|------|--|---|
| (i) | show that $\frac{dy}{dx} = \frac{1}{2}$ at $t = -1$ | 2 |
| (ii) | find the equation of the tangent at the point where $t = -1$ | 1 |
- c) (i) If $P(x) = ax^3 + bx^2 + cx + d$ show that $P(x) - P(\gamma)$ has a factor $(x - \gamma)$. 2
- (ii) Hence show that the polynomial $P(x) - P(\gamma)$ has three distinct roots if 2

$$(b - 3a\gamma)(b + a\gamma) - 4ac > 0$$

Solutions HSC Ext 1 assessment Term 4 2013

$$1. \quad P(4) = 0 \Rightarrow 64a - 7 \times 16 + 4k + 4 = 0 \quad [\div 4]$$

$$16a + k - 27 = 0$$

$$\therefore 16a + k = 27$$

$$P(1) = -6 \Rightarrow a - 7 + k + 4 = -6$$

$$a + k = -3$$

(D)

$$2. \quad ts = -4$$

$$\therefore s = -\frac{4}{t}$$

Midpoint $\left(\frac{2t+2s}{2}, \frac{t^2+s^2}{2}\right) = \left(t+s, \frac{t^2+s^2}{2}\right)$

$$= \left(t - \frac{4}{t}, \frac{t^2 + \frac{16}{t^2}}{2}\right)$$

$$= \left(\frac{t^2-4}{t}, \frac{t^4+16}{2t^2}\right)$$

(B)

$$3. \quad f'(x) = \frac{x^2}{x-2}$$

x	-1	0	1
$f'(x)$	-1/3	0	-1

Horizontal pt of inflexion at $x=0$

(A)

$$4. \quad x^2 = 6y \Rightarrow 4a = 6$$

$$2a = 3$$

Chord of contact $xx_0 = 3(y+y_0)$

$$xx_0 - 3y - 3y_0 = 0$$

(B)

$$\therefore x_0 = 2 \text{ and } y_0 = 1 \text{ to match with } 2x - 3y - 3 = 0$$

$$5. \quad \frac{d}{dx} \left(\frac{x}{\sqrt{1-2x}} \right) = \frac{v u' - u v'}{v^2}$$

$$= \frac{\sqrt{1-2x} - x \times \frac{-1}{\sqrt{1-2x}}}{1-2x}$$

$$u = x \quad v = (1-2x)^{1/2}$$

$$u' = 1 \quad v' = \frac{1}{2}(1-2x)^{-1/2} \left(-2 \right)$$

$$= \frac{(1-2x) + x}{(1-2x)\sqrt{1-2x}} = \frac{1-x}{(1-2x)\sqrt{1-2x}}$$

(A)

$$= -\frac{1}{\sqrt{1-2x}}$$

Question 6

6 a) Roots $1, 2+\sqrt{2}, 2-\sqrt{2}$

$$\begin{aligned}\therefore P(x) &= (x-1)(x^2 - (\alpha+\beta)x + \alpha\beta) \quad \text{where } \alpha+\beta = 2+\sqrt{2} + 2-\sqrt{2} \\ &= 4 \\ &= (x-1)(x^2 - 4x + 2) \\ &= x^3 - 5x^2 + 6x - 2 \\ \alpha\beta &= (2+\sqrt{2})(2-\sqrt{2}) \\ &= 4 - 2 \\ &= 2\end{aligned}$$

6 b) $x^3 - x^2 - 4x = A(x-1)^3 + B(x-1)^2 + C$

Sub $x=1$: $-4 = C$

Coeff of x^3 : $1 = A$

Sub $x=0$: $0 = -A + B + C$

$B = 5$

$$\left. \begin{array}{l} \text{Sub } x=1: -4 = C \\ \text{Coeff of } x^3: 1 = A \\ \text{Sub } x=0: 0 = -A + B + C \end{array} \right\} x^3 - x^2 - 4x = (x-1)^3 + 5(x-1)^2 - 4$$

$b = -5$

$c = 6$

$d = -2$

However this question has an error, because there are other solutions, depending on the method used. ALL possible correctly developed solutions earned FULL marks. Marks were lost for algebraic mistakes.

$$A(x-1)^3 + B(x-1)^2 + C = A(x^3 - 3x^2 + 3x - 1) + B(x^2 - 2x + 1) + C$$

$$\therefore x^3 - x^2 - 4x = Ax^3 - x^2(3A - B) - x(-3A + 2B) - A + B + C$$

Coeff of x^3 : $1 = A$

Coeff of x^2 : $1 = 3A - B$

$\therefore B = 2$

Coeff of x Here is where the conflict is: $-3A + 2B = 4$ doesn't agree

Equating constants: $0 = -A + B + C$

$-1 = C$

The original question for the test was $A(x-1)^3 + B(x-1)^2 + C(x-1) + D$ which has unique solutions $A=1, B=2, C=-3, D=-4$

In our attempt to make the question easier by eliminating the $(x-1)$ term, we introduced an invalid identity.

We apologise for the error. If, ^{in the future,} you believe there's an error in a question, please alert the supervisor at the time.

$$\begin{aligned}
 6c) \quad x &= 2at & y &= at^2 \\
 \frac{dx}{dt} &= 2a & \frac{dy}{dt} &= 2at \\
 \frac{dy}{dx} &= \frac{dy}{dt} \times \frac{dt}{dx} \\
 &= \frac{2at}{2a} \\
 &= t
 \end{aligned}$$

$\therefore$ At $P(2ap, ap^2)$, gradient is p

$$\begin{aligned}
 \text{Tangent: } y - ap^2 &= p(x - 2ap) \\
 y - ap^2 &= px - 2ap^2 \\
 y &= px - ap^2
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) At } T, y=0 &\Rightarrow 0 = px - ap^2 \\
 x &= \frac{ap^2}{p} \\
 &= ap
 \end{aligned}$$

(ii) (cont) $\therefore T$ is $(ap, 0)$

$$\begin{aligned}
 \text{(iii) Normal is } x + py &= 2ap + ap^3 \\
 \text{At } T, x &= ap \\
 \text{At } R, x = ap: & ap + py = 2ap + ap^3 \\
 py &= ap + ap^3 \\
 y &= a + ap^2
 \end{aligned}$$

$\therefore R$ is $(ap, a + ap^2)$

$$\begin{aligned}
 \text{(iv) Let } X &= ap \text{ and } Y = a + ap^2 \\
 \text{Sub } \frac{X}{a} = p &\text{ into } Y = a + a\left(\frac{X}{a}\right)^2 \\
 &= a + \frac{X^2}{a}
 \end{aligned}$$

$$a(Y - a) = X^2$$

$\therefore$ This is a parabola with vertex $(0, a)$ and focal length $\frac{a}{4}$

$$\begin{aligned}
 \text{Directrix} &= a - \frac{a}{4} \\
 &= \frac{3a}{4}
 \end{aligned}$$

Question 7

a) $x^3 - 2x^2 - 7x + 2 = 0$
Possible roots: $\pm 1, \pm 2$

$$P(1) \neq 0, P(-1) \neq 0$$

$$P(2) = 8 - 8 - 14 + 2 \neq 0$$

$$P(-2) = -8 - 8 + 14 + 2 = 0$$

$\therefore$ One root is -2

$$P(x) = (x+2)(x^2 + kx + 1) \text{ by inspection}$$

$$\begin{aligned} P(1) = -6 &\Rightarrow -6 = (1+2)(1+k+1) \\ &= 3(2+k) \\ 2+k &= -2 \\ k &= -4 \end{aligned}$$

$$\therefore \text{Quadratic is } x^2 - 4x + 1 = 0$$

$$\begin{aligned} x &= \frac{4 \pm \sqrt{16 - 4}}{2} \\ &= 2 \pm \sqrt{3} \end{aligned}$$

Alternative solution Having found that one root is -2 then let other roots be α, β

$$\begin{aligned} \alpha + \beta - 2 &= +2 \\ \therefore \alpha + \beta &= 4 \end{aligned}$$

$$\begin{aligned} -2\alpha\beta &= -2 \\ \alpha\beta &= 1 \end{aligned}$$

$$\alpha + \frac{1}{\alpha} = 4$$

$$\begin{aligned} \alpha^2 + 1 &= 4\alpha \\ \alpha^2 - 4\alpha + 1 &= 0 \\ \alpha &= \frac{-4 \pm \sqrt{12}}{2} \\ &= -2 \pm \sqrt{3} \end{aligned}$$

$\therefore$ Roots are $-2, 2+\sqrt{3}, 2-\sqrt{3}$

Alternatively

$$\begin{array}{r} x^2 - 4x + 1 \\ (x+2) \overline{) x^3 - 2x^2 - 7x + 2} \\ \underline{x^3 + 2x^2} \\ -4x^2 - 7x \\ \underline{-4x^2 - 8x} \\ x + 2 \end{array}$$

$$b) P(x) = -2x^3 + kx^2 - mx + 5$$

$$P'(x) = -6x^2 + 2kx - m$$

If $P(x)$ is to have stationary points then

$$-6x^2 + 2kx - m = 0 \text{ must have 1 or 2 solutions}$$

To have solutions, $\Delta \geq 0$

$$\text{i.e. } (2k)^2 - 4(-6)(-m) \geq 0$$

$$4k^2 - 24m \geq 0$$

$$k^2 - 6m \geq 0$$

$$c) y = \frac{(x-1)^2}{x+1}$$

$$(i) \frac{dy}{dx} = \frac{(x+1)2(x-1) - (x-1)^2 \times 1}{(x+1)^2}$$

$$= \frac{2(x^2-1) - (x^2-2x+1)}{(x+1)^2}$$

$$= \frac{x^2 + 2x - 3}{(x+1)^2}$$

$$= \frac{(x+3)(x-1)}{(x+1)^2}$$

Stat pts occur when $\frac{dy}{dx} = 0$, i.e. when $x = -3, 1$

x	-4	-3	-2	0	1	2
$\frac{dy}{dx}$	$\frac{5}{9}$	0	$-\frac{3}{1}$	-3	0	$\frac{5}{9}$

There's a discontinuity at $x = -1$

There is a maximum turning point at $(-3, 8)$ and a minimum turning point at $(1, 0)$

$$\frac{(x-1)^2}{x+1} = \frac{x^2 - 2x + 1}{x+1}$$

$$\begin{array}{r} x-3 \\ x+1 \overline{) x^2 - 2x + 1} \\ \underline{x^2 + x} \\ -3x + 1 \\ \underline{-3x - 3} \\ 4 \end{array}$$

$$\therefore \frac{(x-1)^2}{x+1} = x-3 + \frac{4}{x+1}$$

Alternatively

$$RHS = \frac{(x+1)(x-3) + 4}{x+1}$$

$$= \frac{x^2 - 2x - 3 + 4}{x+1}$$

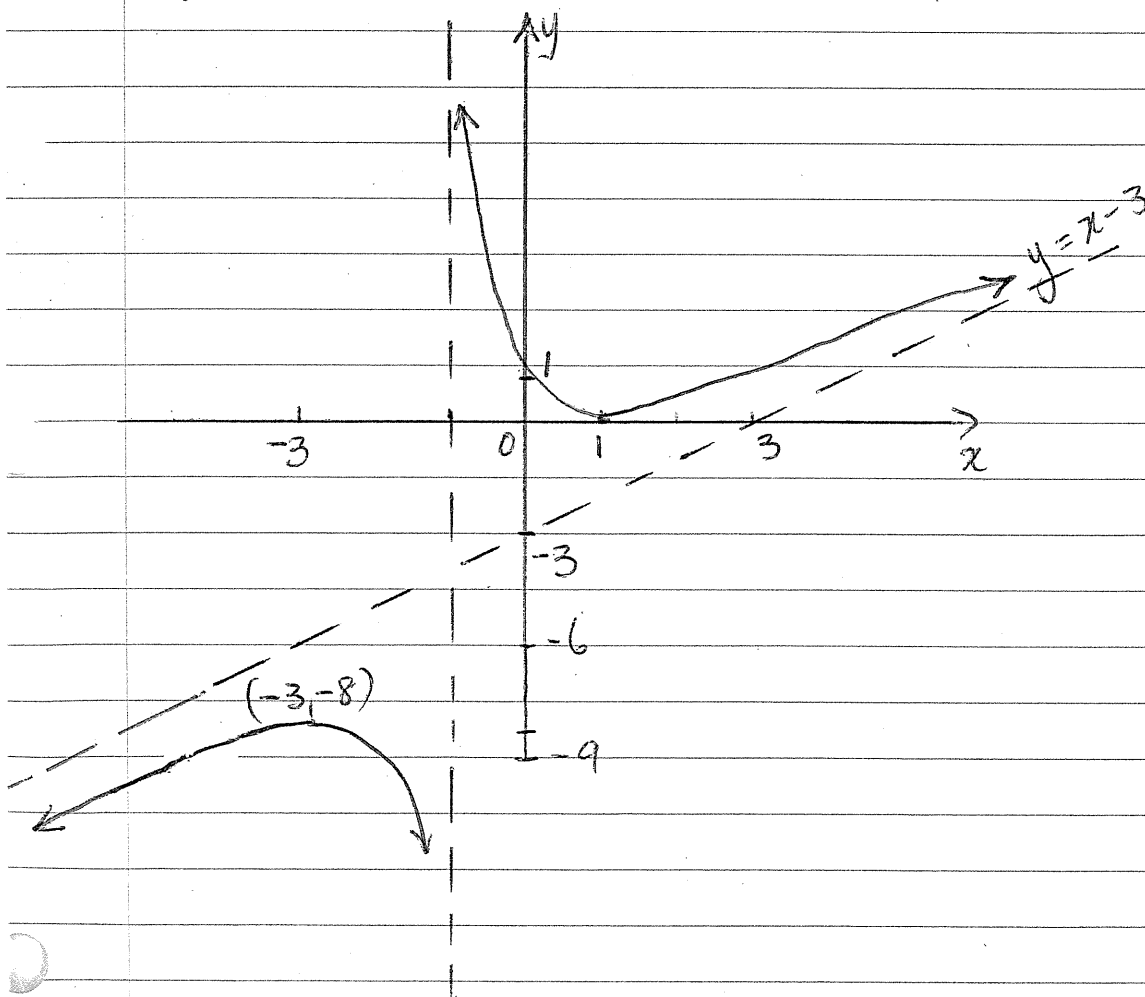
$$= \frac{x^2 - 2x + 1}{x+1}$$

$$= \frac{(x-1)^2}{x+1}$$

$$= LHS$$

(iii) Intercept = (1, 0), (0, 1)

Asymptotes = $x = -1$ and $y = x - 3$



Question 8

(i) Point C (5, 6)

$$(ii) \quad y = \frac{8x^2}{15} - \frac{31x}{15} + 3 \quad 0 \leq x \leq 5$$

$$\frac{dy}{dx} = \frac{16x}{15} - \frac{31}{15}$$

Max gradient will occur when $x=5$ due to ramp being parabolic.

$$\frac{dy}{dx} = \frac{16 \times 5}{15} - \frac{31}{15}$$

$$= \frac{49}{15}$$

$$= 3 \frac{4}{15}$$

$$(iii) \quad \text{Max gradient is } 3 \quad \therefore \quad \frac{16x}{15} - \frac{31}{15} = 3$$

$$x = 4.75$$

$$\therefore y = \frac{8(4.75)^2}{15} - \frac{31 \times 4.75}{15} + 3$$
$$= 5.21 \text{ (2 d.p.)}$$

The level K can be set at 5.2 metres high.

$$b) \quad x = t^2 + 1 \quad \text{and} \quad y = t^{-1}$$

$$\frac{dx}{dt} = 2t$$

$$\frac{dy}{dt} = -\frac{1}{t^2}$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= -\frac{1}{t^2} \times \frac{1}{2t}$$

$$= -\frac{1}{2t^3}$$

$$\text{At } t = -1, \quad \frac{dy}{dx} = +\frac{1}{2}$$

Ex When $t = -1$, $x = 2$ and $y = -1$

Equation of tangent: $y + 1 = \frac{1}{2}(x - 2)$

$$2y + 2 = x - 2$$

$$\therefore x - 2y - 4 = 0$$

c) $P(x) = ax^3 + bx^2 + cx + d$

$$P(x) - P(y) = a(x^3 - y^3) + b(x^2 - y^2) + c(x - y) + d - d$$

$$= a(x - y)(x^2 + xy + y^2) + b(x - y)(x + y) + c(x - y)$$

$$= (x - y) [ax^2 + axy + ay^2 + bx + by + c]$$

$\therefore (x - y)$ is a factor of $P(x)$

(ii) One root will be $x = y$

Other roots will be roots of $(ax^2 + x(ay + b) + ay^2 + by + c) = 0$

If two distinct roots exist to this equation then $\Delta > 0$

$$(ay + b)^2 - 4a(ay^2 + by + c) > 0$$

$$a^2y^2 + 2aby + b^2 - 4a^2y^2 - 4aby - 4ac > 0$$

$$b^2 - 3a^2y^2 - 2aby - 4ac > 0$$

$$(b - 3ay)(b + ay) = b^2 + aby - 3aby - 3a^2y^2$$

$$\therefore (b - 3ay)(b + ay) - 4ac > 0$$